

WINDOWS

```
C:\Users\Анастасия>help
For more information on a specific command, type HELP command-name
ASSOC          Displays or modifies file extension associations.
ATTRIB         Displays or changes file attributes.
BREAK          Sets or clears extended CTRL+C checking.
BCDEDIT        Sets properties in boot database to control boot loading.
CACLS          Displays or modifies access control lists (ACLs) of files.
CALL           Calls one batch program from another.
CD             Displays the name of or changes the current directory.
CHCP           Displays or sets the active code page number.
CHDIR          Displays the name of or changes the current directory.
CHKDSK         Checks a disk and displays a status report.
CHKNTFS        Displays or modifies the checking of disk at boot time.
CLS            Clears the screen.
CMD            Starts a new instance of the Windows command interpreter.
COLOR          Sets the default console foreground and background colors.
COMP           Compares the contents of two files or sets of files.
COMPACT        Displays or alters the compression of files on NTFS partitions.
CONVERT        Converts FAT volumes to NTFS. You cannot convert the
               current drive.
COPY           Copies one or more files to another location.
DATE           Displays or sets the date.
DEL            Deletes one or more files.
DIR            Displays a list of files and subdirectories in a directory.
DISKPART       Displays or configures Disk Partition properties.
DOSKEY         Edits command lines, recalls Windows commands, and
               creates macros.
DRIVERQUERY    Displays current device driver status and properties.
ECHO           Displays messages, or turns command echoing on or off.
ENDLOCAL       Ends localization of environment changes in a batch file.
ERASE          Deletes one or more files.
EXIT           Quits the CMD.EXE program (command interpreter).
FC             Compares two files or sets of files, and displays the
               differences between them.
FIND           Searches for a text string in a file or files.
FINDSTR        Searches for strings in files.
FOR            Runs a specified command for each file in a set of files.
FORMAT         Formats a disk for use with Windows.
FSUTIL         Displays or configures the file system properties.
FTYPE          Displays or modifies file types used in file extension
               associations.
GOTO           Directs the Windows command interpreter to a labeled line in
               a batch program.
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GOTO           Directs the Windows command interpreter to a labeled line in
               a batch program.
GPRESULT       Displays Group Policy information for machine or user.
GRAFTABL       Enables Windows to display an extended character set in
               graphics mode.
HELP           Provides Help information for Windows commands.
ICACLS         Display, modify, backup, or restore ACLs for files and
               directories.
IF             Performs conditional processing in batch programs.
LABEL          Creates, changes, or deletes the volume label of a disk.
MD             Creates a directory.
MKDIR          Creates a directory.
MKLINK         Creates Symbolic Links and Hard Links
MODE           Configures a system device.
MORE           Displays output one screen at a time.
MOVE           Moves one or more files from one directory to another
               directory.
OPENFILES      Displays files opened by remote users for a file share.
PATH           Displays or sets a search path for executable files.
PAUSE          Suspends processing of a batch file and displays a message.
POPD           Restores the previous value of the current directory saved by
               PUSH.D.
PRINT          Prints a text file.
PROMPT         Changes the Windows command prompt.
PUSHD          Saves the current directory then changes it.
RD            Removes a directory.
RECOVER        Recovers readable information from a bad or defective disk.
REM            Records comments (remarks) in batch files or CONFIG.SYS.
REN            Renames a file or files.
RENAME         Renames a file or files.
REPLACE        Replaces files.
RMDIR          Removes a directory.
ROBOCOPY       Advanced utility to copy files and directory trees
SET            Displays, sets, or removes Windows environment variables.
SETLOCAL       Begins localization of environment changes in a batch file.
SC             Displays or configures services (background processes).
SCHTASKS       Schedules commands and programs to run on a computer.
SHIFT          Shifts the position of replaceable parameters in batch files.
SHUTDOWN       Allows proper local or remote shutdown of machine.
SORT           Sorts input.
START          Starts a separate window to run a specified program or command.
SUBST          Associates a path with a drive letter.
SYSTEMINFO     Displays machine specific properties and configuration.
TASKLIST       Displays all currently running tasks including services.
TASKKILL       Kill or stop a running process or application.
TIME           Displays or sets the system time.
TITLE          Sets the window title for a CMD.EXE session.
TREE           Graphically displays the directory structure of a drive or
               path.
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TREE           Graphically displays the directory structure of a drive or
               path.
TYPE           Displays the contents of a text file.
VER            Displays the Windows version.
VERIFY         Tells Windows whether to verify that your files are written
               correctly to a disk.
VOL            Displays a disk volume label and serial number.
XCOPY          Copies files and directory trees.
WMIC           Displays WMI information inside interactive command shell.
```

1. Attrib

```
C:\Users\Анастасия>help attrib
Displays or changes file attributes.

ATTRIB [+R | -R] [+A | -A] [+S | -S] [+H | -H] [+O | -O] [+I | -I] [+X | -X] [+P | -P] [+U | -U]
[drive:][path][filename] [/S [/D]] [/L]

+ Sets an attribute.
- Clears an attribute.
R Read-only file attribute.
A Archive file attribute.
S System file attribute.
H Hidden file attribute.
O Offline attribute.
I Not content indexed file attribute.
X No scrub file attribute.
V Integrity attribute.
P Pinned attribute.
U Unpinned attribute.
B SMR Blob attribute.
[drive:][path][filename]
    Specifies a file or files for attrib to process.
/S Processes matching files in the current folder
    and all subfolders.
/D Processes folders as well.
/L Work on the attributes of the Symbolic Link versus
    the target of the Symbolic Link
```

2. Cacls

```
C:\Users\Анастасия>help cacls

NOTE: Cacls is now deprecated, please use icacls.

Displays or modifies access control lists (ACLs) of files

CACLS filename [/T] [/M] [/L] [/S[:SDDL]] [/E] [/C] [/G user:perm]
[/R user [...]] [/P user:perm [...]] [/D user [...]]
filename      Displays ACLs.
/T            Changes ACLs of specified files in
              the current directory and all subdirectories.
/L            Work on the Symbolic Link itself versus the target
/M            Changes ACLs of volumes mounted to a directory
/S            Displays the SDDL string for the DACL.
/S:SDDL       Replaces the ACLs with those specified in the SDDL string
              (not valid with /E, /G, /R, /P, or /D).
/E            Edit ACL instead of replacing it.
/C            Continue on access denied errors.
/G user:perm  Grant specified user access rights.
              Perm can be: R Read
                          W Write
                          C Change (write)
                          F Full control
/R user       Revoke specified user's access rights (only valid with /E).
/P user:perm  Replace specified user's access rights.
              Perm can be: N None
                          R Read
                          W Write
                          C Change (write)
                          F Full control
/D user       Deny specified user access.
Wildcards can be used to specify more than one file in a command.
You can specify more than one user in a command.

Abbreviations:
CI - Container Inherit.
    The ACE will be inherited by directories.
OI - Object Inherit.
    The ACE will be inherited by files.
IO - Inherit Only.
    The ACE does not apply to the current file/directory.
ID - Inherited.
    The ACE was inherited from the parent directory's ACL.
```

3. Chkdsk

```
C:\Users\Анастасия>help chkdsk
Checks a disk and displays a status report.

CHKDSK [volume[[path]filename]] [/F] [/V] [/R] [/X] [/I] [/C] [/L[:size]] [/B] [/scan] [/spotfix]

volume          Specifies the drive letter (followed by a colon),
                 mount point, or volume name.
filename        FAT/FAT32 only: Specifies the files to check for
                 fragmentation.
/F             Fixes errors on the disk.
/V             On FAT/FAT32: Displays the full path and name of every
                 file on the disk.
                 On NTFS: Displays cleanup messages if any.
/R            Locates bad sectors and recovers readable information
                 (implies /F, when /scan not specified).
/L[:size]      NTFS only: Changes the log file size to the specified
                 number of kilobytes. If size is not specified, displays
                 current size.
/X            Forces the volume to dismount first if necessary.
                 All opened handles to the volume would then be invalid
                 (implies /F).
/I            NTFS only: Performs a less vigorous check of index
                 entries.
/C            NTFS only: Skips checking of cycles within the folder
                 structure.
/B            NTFS only: Re-evaluates bad clusters on the volume
                 (implies /R)
/scan          NTFS only: Runs an online scan on the volume
/forceofflinefix NTFS only: (Must be used with "/scan")
                 Bypass all online repair; all defects found
                 are queued for offline repair (i.e. "chkdsk /spotfix").
/perf          NTFS only: (Must be used with "/scan")
                 Uses more system resources to complete a scan as fast as
                 possible. This may have a negative performance impact on
                 other tasks running on the system.
/spotfix       NTFS only: Runs spot fixing on the volume
/sdcleanup     NTFS only: Garbage collect unneeded security descriptor
                 data (implies /F).
/offlinescanandfix Runs an offline scan and fix on the volume.
/freeorphanedchains FAT/FAT32/exFAT only: Frees any orphaned cluster chains
                 instead of recovering their contents.
/markclean     FAT/FAT32/exFAT only: Marks the volume clean if no
                 corruption was detected, even if /F was not specified.

The /I or /C switch reduces the amount of time required to run Chkdsk by
skipping certain checks of the volume.
```

4. Chkntfs

```
C:\Users\Анастасия>help chkntfs
Displays or modifies the checking of disk at boot time.

CHKNTFS volume [...]
CHKNTFS /D
CHKNTFS /T[:time]
CHKNTFS /X volume [...]
CHKNTFS /C volume [...]

volume          Specifies the drive letter (followed by a colon),
                 mount point, or volume name.
/D             Restores the machine to the default behavior; all drives are
                 checked at boot time and chkdsk is run on those that are
                 dirty.
/T:time        Changes the AUTOCHK initiation countdown time to the
                 specified amount of time in seconds. If time is not
                 specified, displays the current setting.
/X            Excludes a drive from the default boot-time check. Excluded
                 drives are not accumulated between command invocations.
/C            Schedules a drive to be checked at boot time; chkdsk will run
                 if the drive is dirty.

If no switches are specified, CHKNTFS will display if the specified drive is
dirty or scheduled to be checked on next reboot.
```

5. Cmd

```
C:\Users\Анастасия>help cmd
Starts a new instance of the Windows command interpreter

CMD [/A | /U] [/Q] [/D] [/E:ON | /E:OFF] [/F:ON | /F:OFF] [/V:ON | /V:OFF]
[[/S] [/C | /K] string]

/C            Carries out the command specified by string and then terminates
/K            Carries out the command specified by string but remains
/S            Modifies the treatment of string after /C or /K (see below)
/Q            Turns echo off
/D            Disable execution of AutoRun commands from registry (see below)
/A            Causes the output of internal commands to a pipe or file to be ANSI
/U            Causes the output of internal commands to a pipe or file to be
                 Unicode
/T:fg         Sets the foreground/background colors (see COLOR /? for more info)
/E:ON         Enable command extensions (see below)
/E:OFF        Disable command extensions (see below)
/F:ON         Enable file and directory name completion characters (see below)
/F:OFF        Disable file and directory name completion characters (see below)
/V:ON         Enable delayed environment variable expansion using ! as the
                 delimiter. For example, /V:ON would allow !var! to expand the
                 variable var at execution time. The var syntax expands variables
                 at input time, which is quite a different thing when inside of a FOR
                 loop.
/V:OFF        Disable delayed environment expansion.
```

6. Color

Sets the default console foreground and background colors.

COLOR [attr]

attr Specifies color attribute of console output

Color attributes are specified by TWO hex digits -- the first corresponds to the background; the second the foreground. Each digit can be any of the following values:

0 = Black	8 = Gray
1 = Blue	9 = Light Blue
2 = Green	A = Light Green
3 = Aqua	B = Light Aqua
4 = Red	C = Light Red
5 = Purple	D = Light Purple
6 = Yellow	E = Light Yellow
7 = White	F = Bright White

If no argument is given, this command restores the color to what it was when CMD.EXE started. This value either comes from the current console window, the /T command line switch or from the DefaultColor registry value.

The COLOR command sets ERRORLEVEL to 1 if an attempt is made to execute the COLOR command with a foreground and background color that are the same.

Example: "COLOR fc" produces light red on bright white

```
C:\Users\Анастасия>color C
```

```
C:\Users\Анастасия>
```

7. Comp

```
C:\Users\Анастасия>help comp
```

Compares the contents of two files or sets of files.

COMP [data1] [data2] [/D] [/A] [/L] [/N=number] [/C] [/OFF[LINE]] [/M]

data1	Specifies location and name(s) of first file(s) to compare.
data2	Specifies location and name(s) of second files to compare.
/D	Displays differences in decimal format.
/A	Displays differences in ASCII characters.
/L	Displays line numbers for differences.
/N=number	Compares only the first specified number of lines in each file.
/C	Disregards case of ASCII letters when comparing files.
/OFF[LINE]	Do not skip files with offline attribute set.
/M	Do not prompt for compare more files.

To compare sets of files, use wildcards in data1 and data2 parameters.

8. Compact

```

C:\Users\Анастасия>help compact
Displays or alters the compression of files on NTFS partitions.

COMPACT [/C | /U] [/S[:dir]] [/A] [/I] [/F] [/Q] [/EXE[:algorithm]]
        [/CompactOs[:option]] [/WinDir:dir]] [filename [...]]

    /C          Compresses the specified files. Directories will be marked
                  so that files added afterward will be compressed unless /EXE
                  is specified.
    /U          Uncompresses the specified files. Directories will be marked
                  so that files added afterward will not be compressed. If
                  /EXE is specified, only files compressed as executables will
                  be uncompressed; if this is omitted, only NTFS compressed
                  files will be uncompressed.
    /S          Performs the specified operation on files in the given
                  directory and all subdirectories. Default "dir" is the
                  current directory.
    /A          Displays files with the hidden or system attributes. These
                  files are omitted by default.
    /I          Continues performing the specified operation even after errors
                  have occurred. By default, COMPACT stops when an error is
                  encountered.
    /F          Forces the compress operation on all specified files, even
                  those which are already compressed. Already-compressed files
                  are skipped by default.
    /Q          Reports only the most essential information.
    /EXE        Use compression optimized for executable files which are read
                  frequently and not modified. Supported algorithms are:
                  XPRESS4K (fastest) (default)
                  XPRESS8K
                  XPRESS16K
                  LZX      (most compact)
    /CompactOs  Set or query the system's compression state. Supported options are:
                  query - Query the system's Compact state.
                  always - Compress all OS binaries and set the system state to Compact
                           which remains unless administrator changes it.
                  never - Uncompress all OS binaries and set the system state to non
                           Compact which remains unless administrator changes it.
    /WinDir     Used with /CompactOs:query, when querying the offline OS. Specifies
                  the directory where Windows is installed.
    filename    Specifies a pattern, file, or directory.

Used without parameters, COMPACT displays the compression state of
the current directory and any files it contains. You may use multiple
filenames and wildcards. You must put spaces between multiple
parameters.

```

9. Copy

```

C:\Users\Анастасия>help copy
Copies one or more files to another location.

COPY [/D] [/V] [/N] [/Y | /-Y] [/Z] [/L] [/A | /B ] source [/A | /B]
      [+ source [/A | /B] [+ ...]] [destination [/A | /B]]

    source      Specifies the file or files to be copied.
    /A          Indicates an ASCII text file.
    /B          Indicates a binary file.
    /D          Allow the destination file to be created decrypted
    destination Specifies the directory and/or filename for the new file(s).
    /V          Verifies that new files are written correctly.
    /N          Uses short filename, if available, when copying a file with a
                  non-8dot3 name.
    /Y          Suppresses prompting to confirm you want to overwrite an
                  existing destination file.
    /-Y         Causes prompting to confirm you want to overwrite an
                  existing destination file.
    /Z          Copies networked files in restartable mode.
    /L          If the source is a symbolic link, copy the link to the target
                  instead of the actual file the source link points to.

The switch /Y may be preset in the COPYCMD environment variable.
This may be overridden with /-Y on the command line. Default is
to prompt on overwrites unless COPY command is being executed from
within a batch script.

To append files, specify a single file for destination, but multiple files
for source (using wildcards or file1+file2+file3 format).

```

10. Date

```
C:\Users\Анастасия>date
The current date is: 13.02.2024
Enter the new date: (dd-mm-yy) _
```

11. Dir

```
C:\Users\Анастасия>help dir
Displays a list of files and subdirectories in a directory.

DIR [drive:][path][filename] [/A[[:]attributes]] [/B] [/C] [/D] [/L] [/N]
  [/O[[:]sortorder]] [/P] [/Q] [/R] [/S] [/T[[:]timefield]] [/W] [/X] [/4]

  [drive:][path][filename]
      Specifies drive, directory, and/or files to list.

  /A      Displays files with specified attributes.
  attributes  D Directories          R Read-only files
               H Hidden files         A Files ready for archiving
               S System files         I Not content indexed files
               L Reparse Points       O Offline files
               - Prefix meaning not

  /B      Uses bare format (no heading information or summary).
  /C      Display the thousand separator in file sizes. This is the
           default. Use /-C to disable display of separator.
  /D      Same as wide but files are list sorted by column.
  /L      Uses lowercase.
  /N      New long list format where filenames are on the far right.
  /O      List by files in sorted order.
  sortorder  N By name (alphabetic)      S By size (smallest first)
               E By extension (alphabetic) D By date/time (oldest first)
               G Group directories first - Prefix to reverse order

  /P      Pauses after each screenful of information.
  /Q      Display the owner of the file.
  /R      Display alternate data streams of the file.
  /S      Displays files in specified directory and all subdirectories.
  /T      Controls which time field displayed or used for sorting
  timefield  C Creation
               A Last Access
               W Last Written

  /W      Uses wide list format.
  /X      This displays the short names generated for non-8dot3 file
           names. The format is that of /N with the short name inserted
           before the long name. If no short name is present, blanks are
           displayed in its place.
  /4      Displays four-digit years

Switches may be preset in the DIRCMD environment variable. Override
preset switches by prefixing any switch with - (hyphen)--for example, /-W.
```

12. Help

```
C:\Users\Анастасия>help
For more information on a specific command, type HELP command-name
ASSOC          Displays or modifies file extension associations.
ATTRIB         Displays or changes file attributes.
BREAK          Sets or clears extended CTRL+C checking.
BCDEDIT        Sets properties in boot database to control boot loading.
CACLS          Displays or modifies access control lists (ACLs) of files.
CALL           Calls one batch program from another.
CD             Displays the name of or changes the current directory.
CHCP           Displays or sets the active code page number.
CHDIR          Displays the name of or changes the current directory.
CHKDSK         Checks a disk and displays a status report.
CHKNTFS        Displays or modifies the checking of disk at boot time.
CLS           Clears the screen.
```

13. Time

```
C:\Users\Анастасия>time
The current time is: 16:06:48,75
Enter the new time:
```

14. Tree

```
C:\Users\Анастасия>help tree
Graphically displays the folder structure of a drive or path.

TREE [drive:][path] [/F] [/A]

    /F  Display the names of the files in each folder.
    /A  Use ASCII instead of extended characters.
```

15. Print

```
C:\Users\Анастасия>help print
Prints a text file.

PRINT [/D:device] [[drive:][path]filename[...]]

    /D:device  Specifies a print device.
```

16. robocopy

```

C:\Users\Анастасия>help robocopy

-----
ROBOCOPY      ::      Robust File Copy for Windows
-----

Started : 13 февраля 2024 г. 15:58:57
Usage :: ROBOCOPY source destination [file [file]...] [options]

    source :: Source Directory (drive:\path or \\server\share\path).
    destination :: Destination Dir (drive:\path or \\server\share\path).
    file :: File(s) to copy (names/wildcards: default is "*.*").

::
:: Copy options :
::
    /S :: copy Subdirectories, but not empty ones.
    /E :: copy subdirectories, including Empty ones.
    /LEV:n :: only copy the top n LEVels of the source directory tree.

    /Z :: copy files in restartable mode.
    /B :: copy files in Backup mode.
    /ZB :: use restartable mode; if access denied use Backup mode.
    /J :: copy using unbuffered I/O (recommended for large files).
    /EFSRAW :: copy all encrypted files in EFS RAW mode.

    /COPY:copyflag[s] :: what to COPY for files (default is /COPY:DAT).
                        (copyflags : D=Data, A=Attributes, T=Timestamps, X=Skip alt data streams).
                        (S=Security=NTFS ACLs, O=Owner info, U=auditing info).

```

17. subst

```

C:\Users\Анастасия>help subst
Associates a path with a drive letter.

SUBST [drive1: [drive2:]path]
SUBST drive1: /D

    drive1:          Specifies a virtual drive to which you want to assign a path.
    [drive2:]path    Specifies a physical drive and path you want to assign to
                     a virtual drive.

    /D               Deletes a substituted (virtual) drive.

Type SUBST with no parameters to display a list of current virtual drives.

```

18. shutdown

```

C:\Users\Анастасия>help shutdown
Usage: shutdown [/i | /l | /s | /sg | /r | /g | /a | /p | /h | /e | /o] [/hybrid] [/soft] [/fw] [/f]
        [/m \\computer][/t xxx][/d [p|u:]xx:yy [/c "comment"]]

No args    Display help. This is the same as typing /?.
/?         Display help. This is the same as not typing any options.
/i         Display the graphical user interface (GUI).
           This must be the first option.
/l         Log off. This cannot be used with /m or /d options.
/s         Shutdown the computer.
/sg        Shutdown the computer. On the next boot, if Automatic Restart Sign-On
           is enabled, automatically sign in and lock last interactive user.
           After sign in, restart any registered applications.
/r         Full shutdown and restart the computer.
/g         Full shutdown and restart the computer. After the system is rebooted,
           if Automatic Restart Sign-On is enabled, automatically sign in and
           lock last interactive user.
           After sign in, restart any registered applications.
/a         Abort a system shutdown.
           This can only be used during the time-out period.
           Combine with /fw to clear any pending boots to firmware.

```

19. move


```

C:\Users\Анастасия> help move
Moves files and renames files and directories.

To move one or more files:
MOVE [/Y | /-Y] [drive:][path]filename1[,...] destination

To rename a directory:
MOVE [/Y | /-Y] [drive:][path]dirname1 dirname2

[drive:][path]filename1 Specifies the location and name of the file
                        or files you want to move.
destination             Specifies the new location of the file. Destination
                        can consist of a drive letter and colon, a
                        directory name, or a combination. If you are moving
                        only one file, you can also include a filename if
                        you want to rename the file when you move it.
[drive:][path]dirname1 Specifies the directory you want to rename.
dirname2                Specifies the new name of the directory.

/Y                      Suppresses prompting to confirm you want to
                        overwrite an existing destination file.
/-Y                     Causes prompting to confirm you want to overwrite
                        an existing destination file.

The switch /Y may be present in the COPYCMD environment variable.
This may be overridden with /-Y on the command line. Default is
to prompt on overwrites unless MOVE command is being executed from
within a batch script.

```

20. find

```

C:\Users\Анастасия>help find
Searches for a text string in a file or files.

FIND [/V] [/C] [/N] [/I] [/OFF[LINE]] "string" [[drive:][path]filename[ ...]]

/V          Displays all lines NOT containing the specified string.
/C          Displays only the count of lines containing the string.
/N          Displays line numbers with the displayed lines.
/I          Ignores the case of characters when searching for the string.
/OFF[LINE] Do not skip files with offline attribute set.
"string"    Specifies the text string to find.
[drive:][path]filename
            Specifies a file or files to search.

If a path is not specified, FIND searches the text typed at the prompt
or piped from another command.

```

LINUX

```

job_spec [&]
(( expression ))
. filename [arguments]
:
[ arg... ]
[[ expression ]]
alias [-p] [name[=value] ... ]
bg [job_spec ...]
bind [-lpsvPSVX] [-m keymap] [-f file]
break [n]
builtin [shell-builtin [arg ...]]
caller [expr]
case WORD in [PATTERN [| PATTERN]...)>
cd [-L][-P [-e]] [-@]] [dir]
command [-pVv] command [arg ...]
compgen [-abdefgjkshuv] [-o option] [>
complete [-abdefgjkshuv] [-pr] [-DEI]>
comptop [-o|+o option] [-DEI] [name .>
continue [n]
coproc [NAME] command [redirections]
declare [-aAfFgiIlNrtux] [-p] [name[=>
dirs [-clpv] [+N] [-N]
disown [-h] [-ar] [jobspec ... | pid >
echo [-neE] [arg ...]
enable [-a] [-dnps] [-f filename] [na>
eval [arg ...]
exec [-cl] [-a name] [command [argume>
exit [n]
history [-c] [-d offset] [n] or hist>
if COMMANDS; then COMMANDS; [ elif C>
jobs [-lnprs] [jobspec ...] or jobs >
kill [-s sigspec | -n signum | -sigs>
let arg [arg ...]
local [option] name[=value] ...
logout [n]
mapfile [-d delim] [-n count] [-O or>
popd [-n] [+N | -N]
printf [-v var] format [arguments]
pushd [-n] [+N | -N | dir]
pwd [-LP]
read [-ers] [-a array] [-d delim] [->
readarray [-d delim] [-n count] [-O >
readonly [-aAf] [name[=value] ...] o>
return [n]
select NAME [in WORDS ... ;] do COMM>
set [-abefhkmnptuvxBCHP] [-o option->
shift [n]
shopt [-pqsu] [-o] [optname ...]
source filename [arguments]
suspend [-f]
test [expr]
time [-p] pipeline
times
trap [-lp] [[arg] signal_spec ...]
true
type [-afptP] name [name ...]

export [-fn] [name[=value] ...] or ex>
false
fc [-e ename] [-lnr] [first] [last] o>
fg [job_spec]
for NAME [in WORDS ... ] ; do COMMAND>
for (( exp1; exp2; exp3 )); do COMMAN>
function name { COMMANDS ; } or name >
getopts optstring name [arg ...]
hash [-lr] [-p pathname] [-dt] [name >
help [-dms] [pattern ...]
typeset [-aAfFgiIlNrtux] [-p] name[=>
ulimit [-SHabdefiklmnpqrstuvxPT] [l>
umask [-p] [-S] [mode]
unalias [-a] name [name ...]
unset [-f] [-v] [-n] [name ...]
until COMMANDS; do COMMANDS; done
variables - Names and meanings of so>
wait [-fn] [-p var] [id ...]
while COMMANDS; do COMMANDS; done
{ COMMANDS ; }

```

1. cd

```
$ help cd
cd: cd [-L|[-P [-e]] [-@]] [dir]
    Change the shell working directory.

    Change the current directory to DIR.  The default DIR is the value of the
    HOME shell variable.

    The variable CDPATH defines the search path for the directory containing
    DIR.  Alternative directory names in CDPATH are separated by a colon (:).
    A null directory name is the same as the current directory.  If DIR begins
    with a slash (/), then CDPATH is not used.

    If the directory is not found, and the shell option `cdable_vars' is set,
    the word is assumed to be a variable name.  If that variable has a value,
    its value is used for DIR.

Options:
  -L    force symbolic links to be followed: resolve symbolic
        links in DIR after processing instances of `..'
  -P    use the physical directory structure without following
        symbolic links: resolve symbolic links in DIR before
        processing instances of `..'
  -e    if the -P option is supplied, and the current working
        directory cannot be determined successfully, exit with
        a non-zero status
  -@    on systems that support it, present a file with extended
        attributes as a directory containing the file attributes
```

2. command

```
$ help command
export "PS1=$ "
help command

command: command [-pVv] command [arg ...]
    Execute a simple command or display information about commands.

Runs COMMAND with ARGS suppressing shell function lookup, or display
information about the specified COMMANDs. Can be used to invoke commands
on disk when a function with the same name exists.

Options:
  -p    use a default value for PATH that is guaranteed to find all of
        the standard utilities
  -v    print a description of COMMAND similar to the `type' builtin
  -V    print a more verbose description of each COMMAND

Exit Status:
    Returns exit status of COMMAND, or failure if COMMAND is not found.
export "PS1=$ "
```

3. dirs

```
help dirs
```

```
dirs: dirs [-clpv] [+N] [-N]
```

Display directory stack.

Display the list of currently remembered directories. Directories find their way onto the list with the ``pushd'` command; you can get back up through the list with the ``popd'` command.

Options:

- | -c clear the directory stack by deleting all of the elements
- | -l do not print tilde-prefixed versions of directories relative to your home directory
- | -p print the directory stack with one entry per line
- | -v print the directory stack with one entry per line prefixed with its position in the stack

Arguments:

- | +N Displays the Nth entry counting from the left of the list shown by `dirs` when invoked without options, starting with zero.
- | -N Displays the Nth entry counting from the right of the list shown by `dirs` when invoked without options, starting with zero.

4. echo

```
help echo
```

```
echo: echo [-neE] [arg ...]
```

Write arguments to the standard output.

Display the ARGs, separated by a single space character and followed by a newline, on the standard output.

Options:

- n do not append a newline
- e enable interpretation of the following backslash escapes
- E explicitly suppress interpretation of backslash escapes

'echo' interprets the following backslash-escaped characters:

- \a alert (bell)
- \b backspace
- \c suppress further output
- \e escape character
- \E escape character
- \f form feed
- \n new line
- \r carriage return
- \t horizontal tab
- \v vertical tab
- \\ backslash

5. eval

```
$ help eval
```

```
export "PS1=$ "
```

```
help eval
```

```
eval: eval [arg ...]
```

Execute arguments as a shell command.

Combine ARGs into a single string, use the result as input to the shell, and execute the resulting commands.

Exit Status:

Returns exit status of command or success if command is null.

```
export "PS1=$ "
```

6. exec

```
help exec
```

```
exec: exec [-cl] [-a name] [command [argument ...]] [redirection ...]
```

Replace the shell with the given command.

Execute COMMAND, replacing this shell with the specified program.

ARGUMENTS become the arguments to COMMAND. If COMMAND is not specified, any redirections take effect in the current shell.

Options:

-a name pass NAME as the zeroth argument to COMMAND

-c execute COMMAND with an empty environment

-l place a dash in the zeroth argument to COMMAND

If the command cannot be executed, a non-interactive shell exits, unless the shell option `execfail' is set.

Exit Status:

Returns success unless COMMAND is not found or a redirection error occurs.

```
export "PS1=$ "
```

7. kill

```
help kill
```

```
kill: kill [-s sigspec | -n signum | -sigspec] pid | jobspec ... or kill -l [sigspec]
```

Send a signal to a job.

Send the processes identified by PID or JOBSPEC the signal named by SIGSPEC or SIGNUM. If neither SIGSPEC nor SIGNUM is present, then SIGTERM is assumed.

Options:

-s sig SIG is a signal name

-n sig SIG is a signal number

-l list the signal names; if arguments follow `-l' they are assumed to be signal numbers for which names should be listed

-L synonym for -l

Kill is a shell builtin for two reasons: it allows job IDs to be used instead of process IDs, and allows processes to be killed if the limit on processes that you can create is reached.

Exit Status:

Returns success unless an invalid option is given or an error occurs.

```
export "PS1=$ "
```

8. local

```
help local
```

```
local: local [option] name[=value] ...
```

```
Define local variables.
```

```
Create a local variable called NAME, and give it VALUE.  OPTION can  
be any option accepted by `declare'.
```

```
Local variables can only be used within a function; they are visible  
only to the function where they are defined and its children.
```

```
Exit Status:
```

```
Returns success unless an invalid option is supplied, a variable  
assignment error occurs, or the shell is not executing a function.
```

```
export "PS1=$ "
```

9. popd


```
help popd
```

```
popd: popd [-n] [+N | -N]
```

Remove directories from stack.

Removes entries from the directory stack. With no arguments, removes the top directory from the stack, and changes to the new top directory.

Options:

- n Suppresses the normal change of directory when removing directories from the stack, so only the stack is manipulated.

Arguments:

- +N Removes the Nth entry counting from the left of the list shown by `dirs`, starting with zero. For example: `popd +0` removes the first directory, `popd +1` the second.

- N Removes the Nth entry counting from the right of the list shown by `dirs`, starting with zero. For example: `popd -0` removes the last directory, `popd -1` the next to last.

The `dirs` builtin displays the directory stack.

Exit Status:

Returns success unless an invalid argument is supplied or the directory change fails.

```
export "PS1=$ "
```

10. printf

```
help printf
```

```
printf: printf [-v var] format [arguments]
```

```
Formats and prints ARGUMENTS under control of the FORMAT.
```

```
Options:
```

```
-v var    assign the output to shell variable VAR rather than  
          display it on the standard output
```

FORMAT is a character string which contains three types of objects: plain characters, which are simply copied to standard output; character escape sequences, which are converted and copied to the standard output; and format specifications, each of which causes printing of the next successive argument.

In addition to the standard format specifications described in printf(1), printf interprets:

```
%b    expand backslash escape sequences in the corresponding argument  
%q    quote the argument in a way that can be reused as shell input  
%(fmt)T  output the date-time string resulting from using FMT as a format  
          string for strftime(3)
```

The format is re-used as necessary to consume all of the arguments. If there are fewer arguments than the format requires, extra format specifications behave as if a zero value or null string, as appropriate, had been supplied.

11. pwd

```
help pwd
```

```
pwd: pwd [-LP]
```

Print the name of the current working directory.

Options:

- L print the value of \$PWD if it names the current working directory
- P print the physical directory, without any symbolic links

By default, `pwd` behaves as if `-L` were specified.

Exit Status:

Returns 0 unless an invalid option is given or the current directory cannot be read.

```
export "PS1=$ "
```

12. read

```
help read
```

```
read: read [-ers] [-a array] [-d delim] [-i text] [-n nchars] [-N nchars] [-p prompt] [-t timeout] [-u fd] [name ...]
```

Read a line from the standard input and split it into fields.

Reads a single line from the standard input, or from file descriptor FD if the -u option is supplied. The line is split into fields as with word splitting, and the first word is assigned to the first NAME, the second word to the second NAME, and so on, with any leftover words assigned to the last NAME. Only the characters found in \$IFS are recognized as word delimiters.

If no NAMES are supplied, the line read is stored in the REPLY variable.

Options:

- a array assign the words read to sequential indices of the array variable ARRAY, starting at zero
- d delim continue until the first character of DELIM is read, rather than newline
- e use Readline to obtain the line
- i text use TEXT as the initial text for Readline
- n nchars return after reading NCHARS characters rather than waiting for a newline, but honor a delimiter if fewer than NCHARS characters are read before the delimiter

13. type

```
help type
```

```
type: type [-afptP] name [name ...]
```

Display information about command type.

For each NAME, indicate how it would be interpreted if used as a command name.

Options:

- a display all locations containing an executable named NAME; includes aliases, builtins, and functions, if and only if the `-p` option is not also used
- f suppress shell function lookup
- P force a PATH search for each NAME, even if it is an alias, builtin, or function, and returns the name of the disk file that would be executed
- p returns either the name of the disk file that would be executed, or nothing if ``type -t NAME'` would not return ``file'`
- t output a single word which is one of ``alias'`, ``keyword'`, ``function'`, ``builtin'`, ``file'` or ``'`, if NAME is an alias, shell reserved word, shell function, shell builtin, disk file, or not found, respectively

14. set

```
help set
```

```
set: set [-abefhkmnptuvxBCHP] [-o option-name] [--] [arg ...]
```

Set or unset values of shell options and positional parameters.

Change the value of shell attributes and positional parameters, or display the names and values of shell variables.

Options:

- a Mark variables which are modified or created for export.
- b Notify of job termination immediately.
- e Exit immediately if a command exits with a non-zero status.
- f Disable file name generation (globbing).
- h Remember the location of commands as they are looked up.
- k All assignment arguments are placed in the environment for a command, not just those that precede the command name.
- m Job control is enabled.
- n Read commands but do not execute them.

15. shopt

```
help shopt
```

```
shopt: shopt [-pqsu] [-o] [optname ...]
```

Set and unset shell options.

Change the setting of each shell option OPTNAME. Without any option arguments, list each supplied OPTNAME, or all shell options if no OPTNAMEs are given, with an indication of whether or not each is set.

Options:

- o restrict OPTNAMEs to those defined for use with `set -o`
- p print each shell option with an indication of its status
- q suppress output
- s enable (set) each OPTNAME
- u disable (unset) each OPTNAME

16. time

```
help time
```

```
time: time [-p] pipeline
```

Report time consumed by pipeline's execution.

Execute PIPELINE and print a summary of the real time, user CPU time, and system CPU time spent executing PIPELINE when it terminates.

Options:

- p print the timing summary in the portable Posix format

The value of the TIMEFORMAT variable is used as the output format.

17. help

```
help help
```

```
help: help [-dms] [pattern ...]
```

Display information about builtin commands.

Displays brief summaries of builtin commands. If PATTERN is specified, gives detailed help on all commands matching PATTERN, otherwise the list of help topics is printed.

Options:

- d output short description for each topic
- m display usage in pseudo-manpage format
- s output only a short usage synopsis for each topic matching PATTERN

18. unset

```
help unset
```

```
unset: unset [-f] [-v] [-n] [name ...]
```

Unset values and attributes of shell variables and functions.

For each NAME, remove the corresponding variable or function.

Options:

- f treat each NAME as a shell function
- v treat each NAME as a shell variable
- n treat each NAME as a name reference and unset the variable itself rather than the variable it references

19. umask

```
help umask
```

```
umask: umask [-p] [-S] [mode]
```

Display or set file mode mask.

Sets the user file-creation mask to MODE. If MODE is omitted, prints the current value of the mask.

If MODE begins with a digit, it is interpreted as an octal number; otherwise it is a symbolic mode string like that accepted by chmod(1).

Options:

- p if MODE is omitted, output in a form that may be reused as input
- S makes the output symbolic; otherwise an octal number is output

20. declare

```
help declare
```

```
declare: declare [-aAfFgiIlNrTux] [-p] [name[=value] ...]
```

Set variable values and attributes.

Declare variables and give them attributes. If no NAMES are given, display the attributes and values of all variables.

Options:

- f restrict action or display to function names and definitions
- F restrict display to function names only (plus line number and source file when debugging)
- g create global variables when used in a shell function; otherwise ignored
- I if creating a local variable, inherit the attributes and value of a variable with the same name at a previous scope
- p display the attributes and value of each NAME